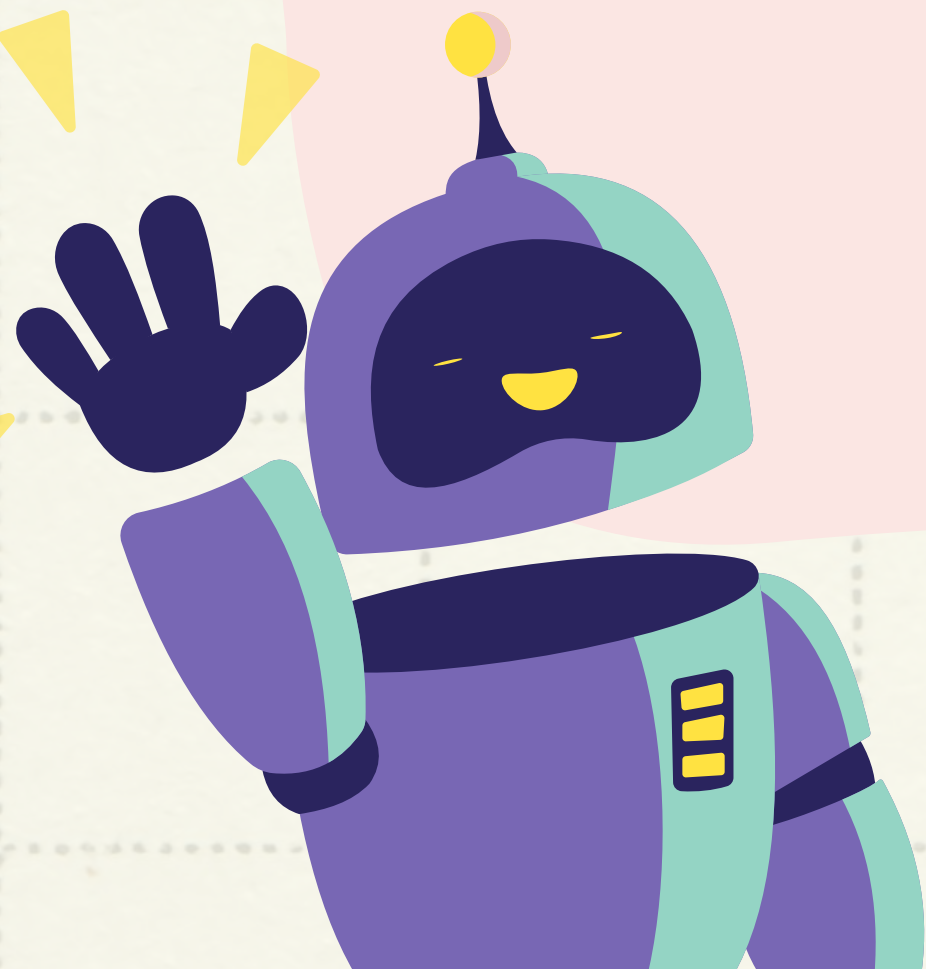




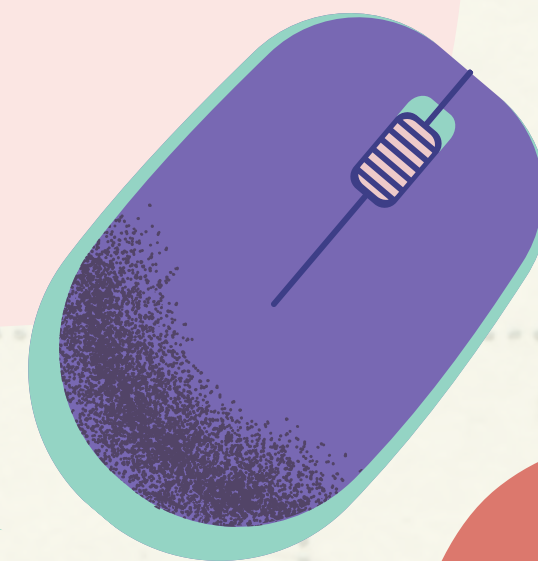
2D DESIGN AND CUTTING

PS70 LAB

llll

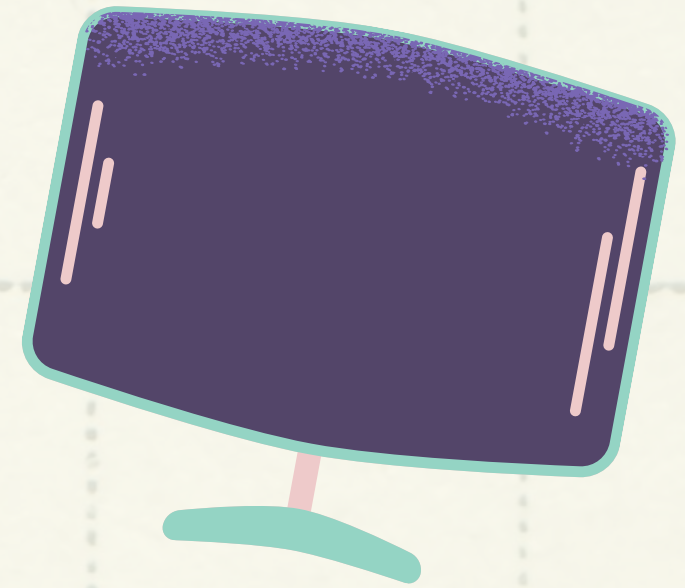


llll



Reminder: Install
Fusion if you
haven't already.

TODAY'S LAB



Vinyl Cutter Demo

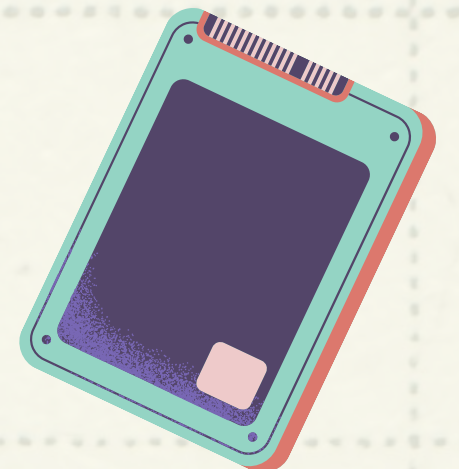
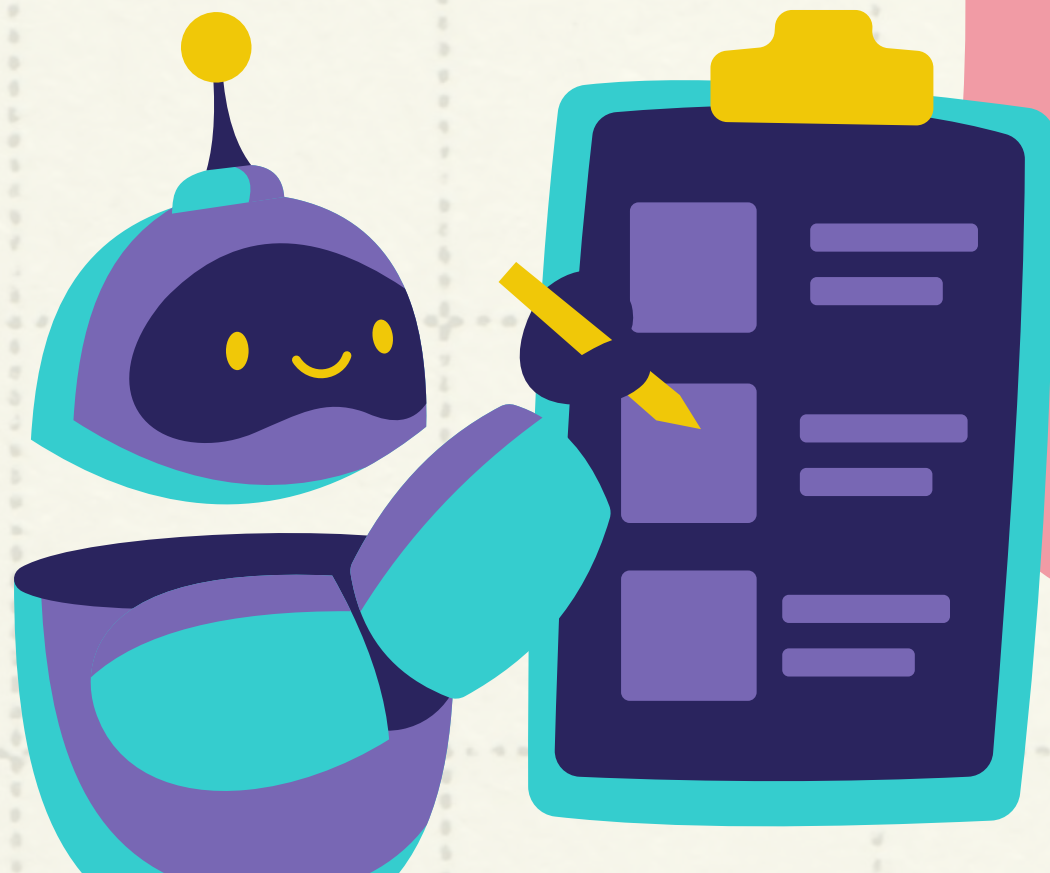
Useless Box: Design and Cut Box
Walls Out of Cardboard

Work on the Assignment:
Create Your Materials Box

Fusion Tutorials

3D Model 2 Objects

Cut Something on the Vinyl Cutter





1

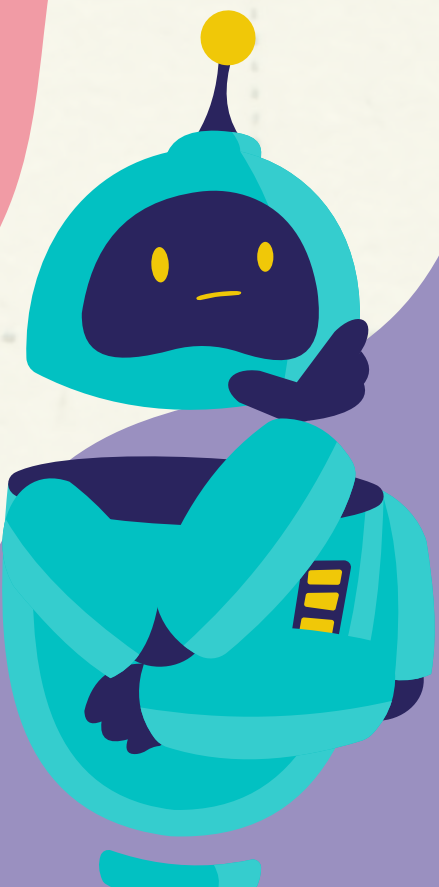
VINYL CUTTER



2

LAB PROJECT:

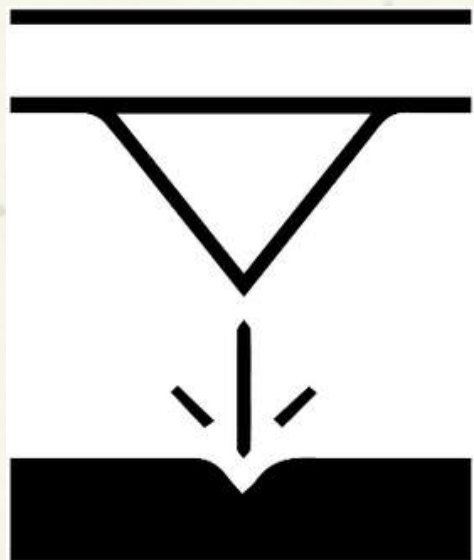
BUILD A USELESS BOX



OVER THE NEXT FEW WEEKS WE WILL BUILD

A USELESS BOX

Week 2



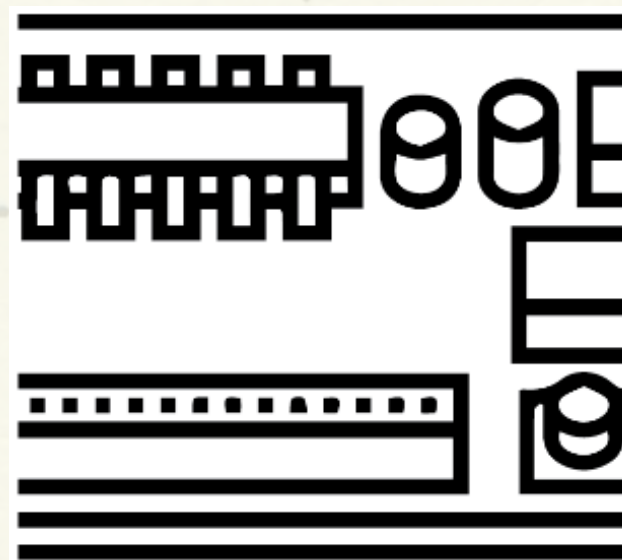
Design and 2D
Cut Box Walls

Week 3



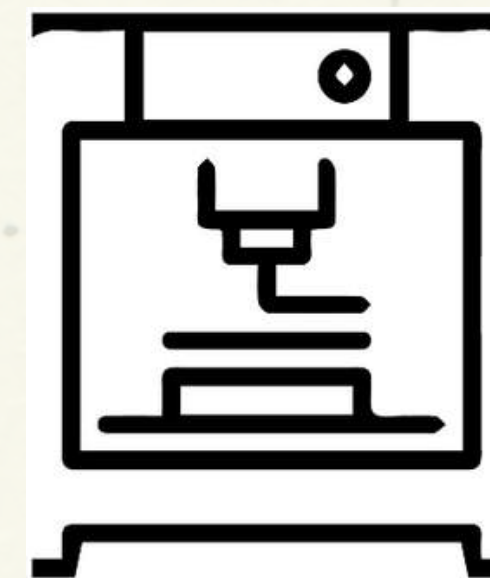
Make Bottom
of the Box with
Handtools

Week 4



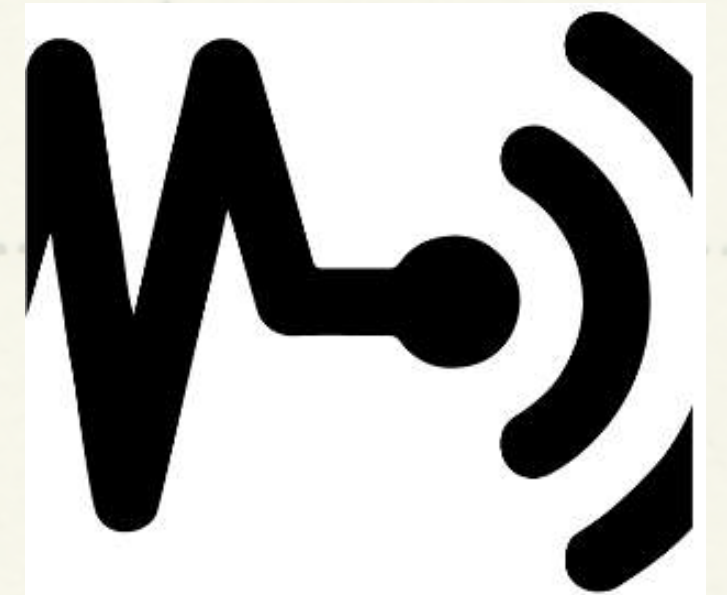
Program the
Servos

Week 5

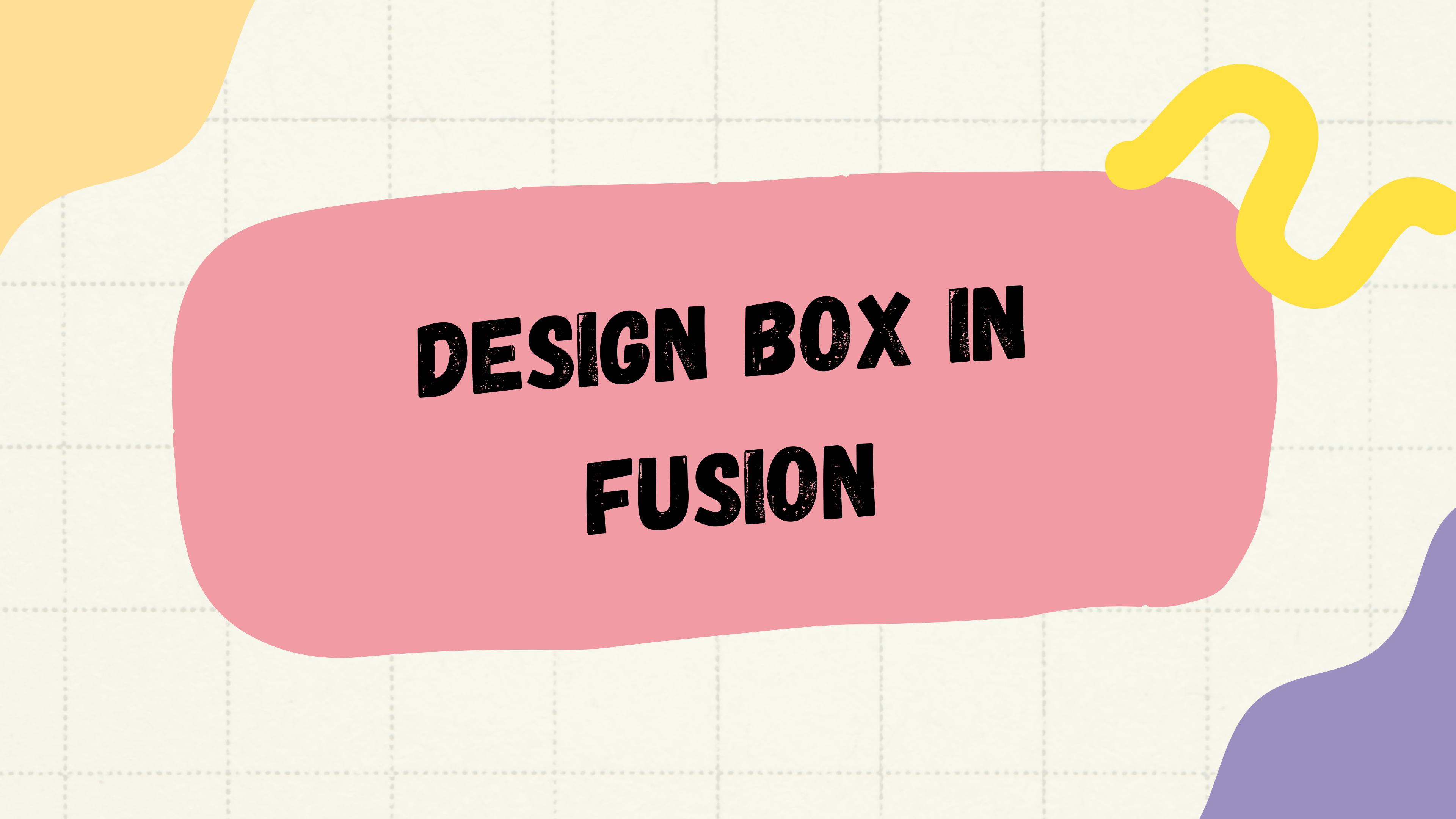


3D Print the
Servo Holder

Week 6

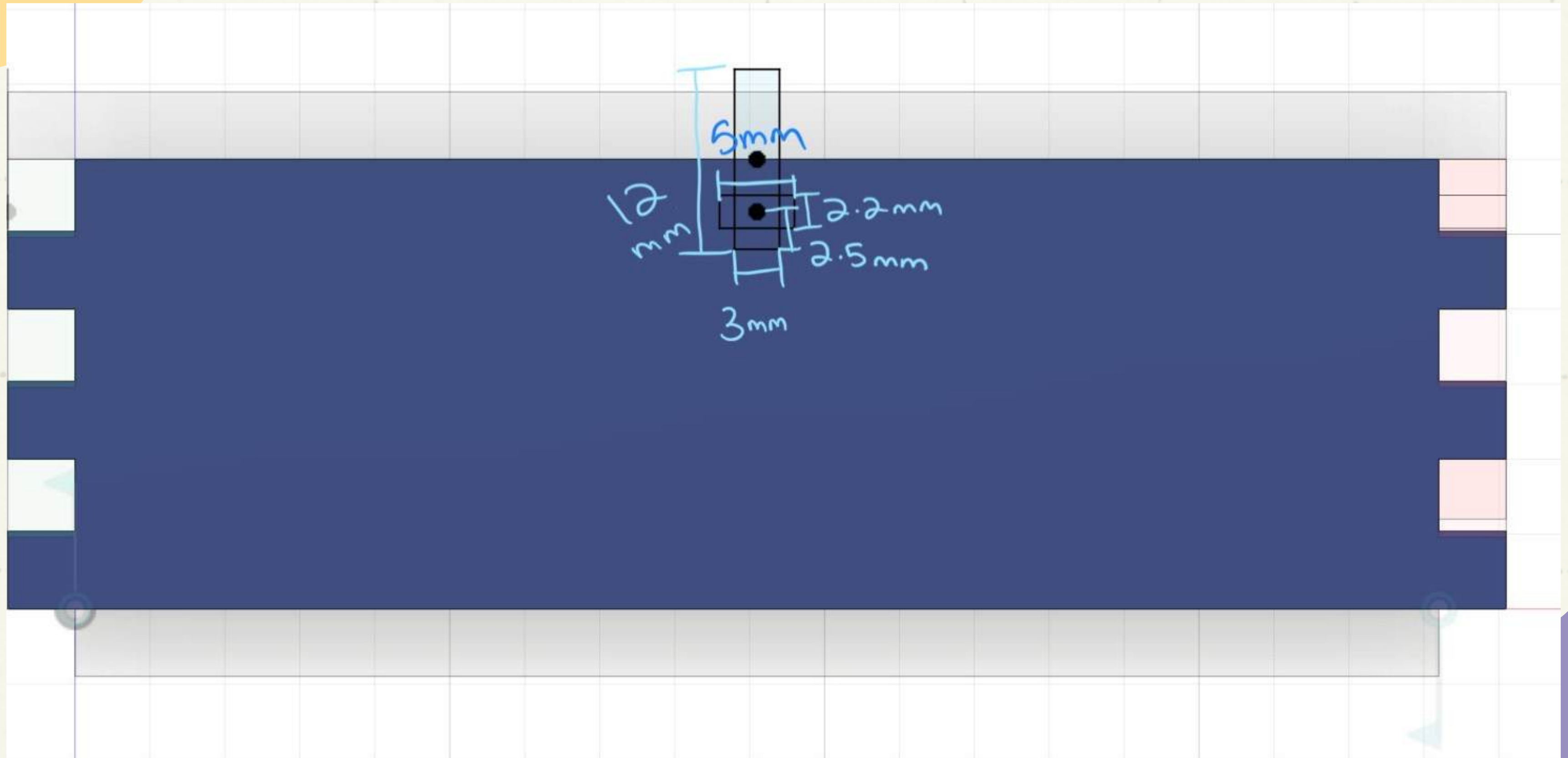


Program a
Switch to Open
the Box



DESIGN BOX IN FUSION

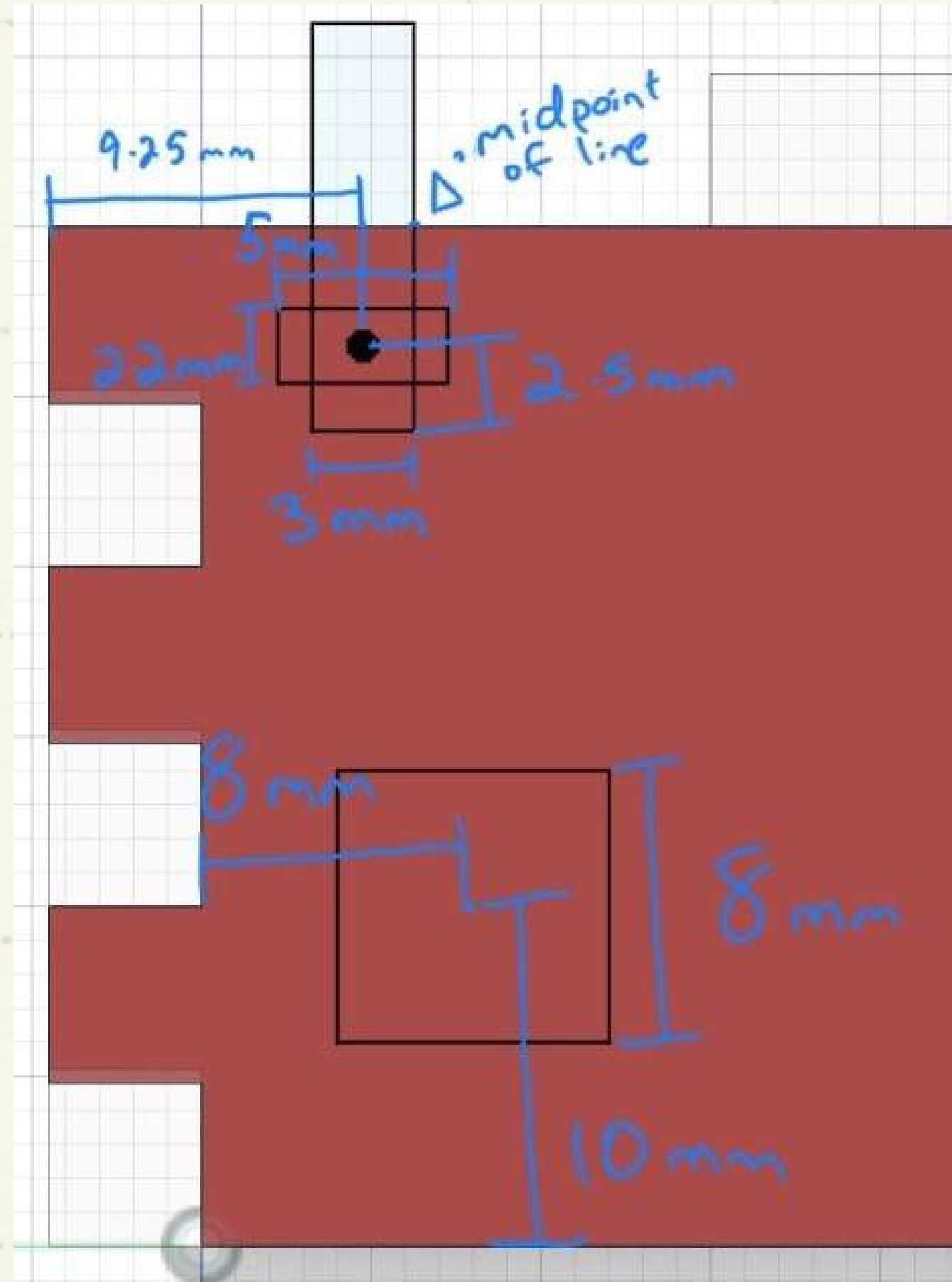
DESIGN BOX IN FUSION360: FRONT PANEL



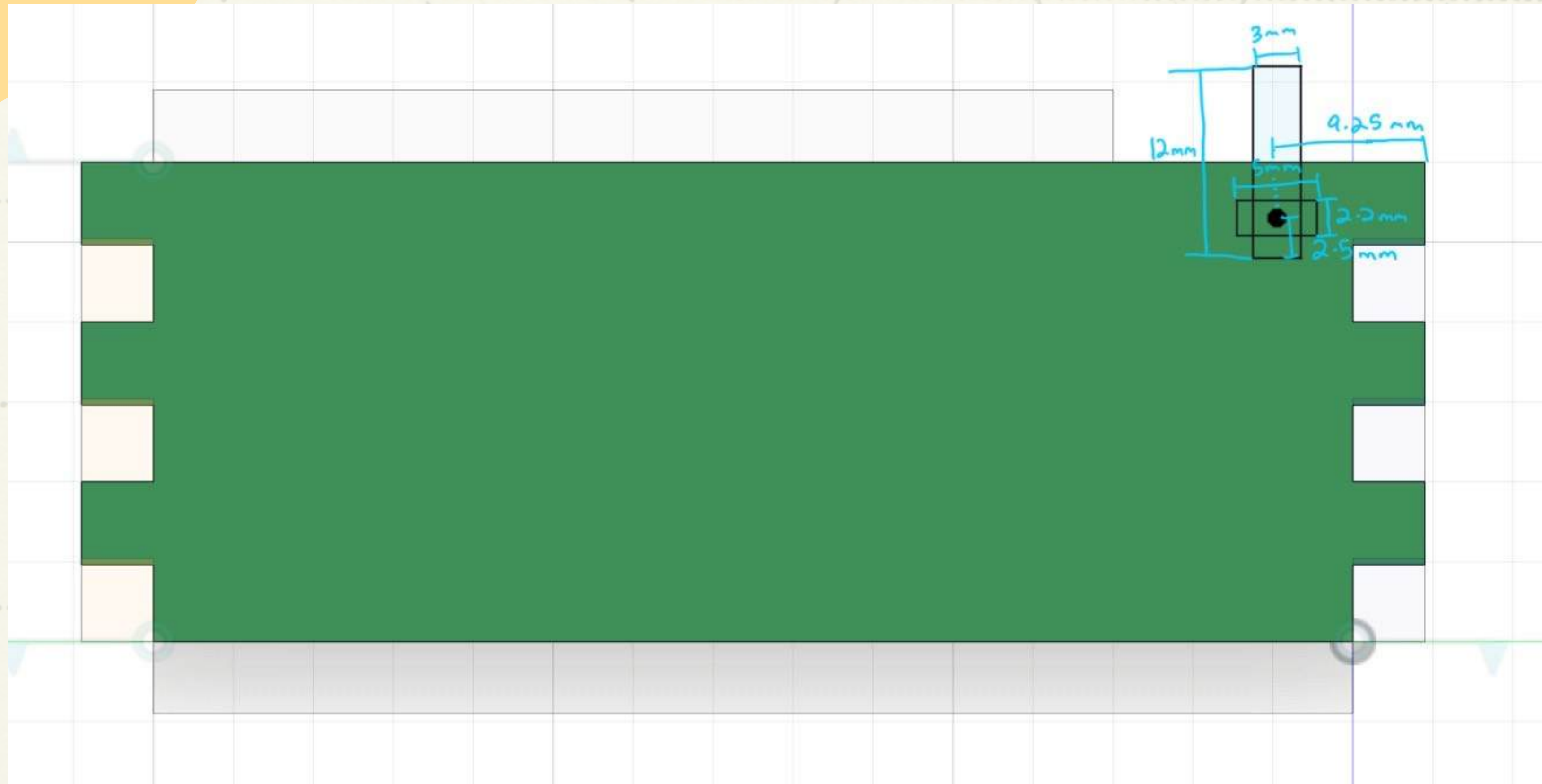
DESIGN BOX IN FUSION360: LEFT PANEL



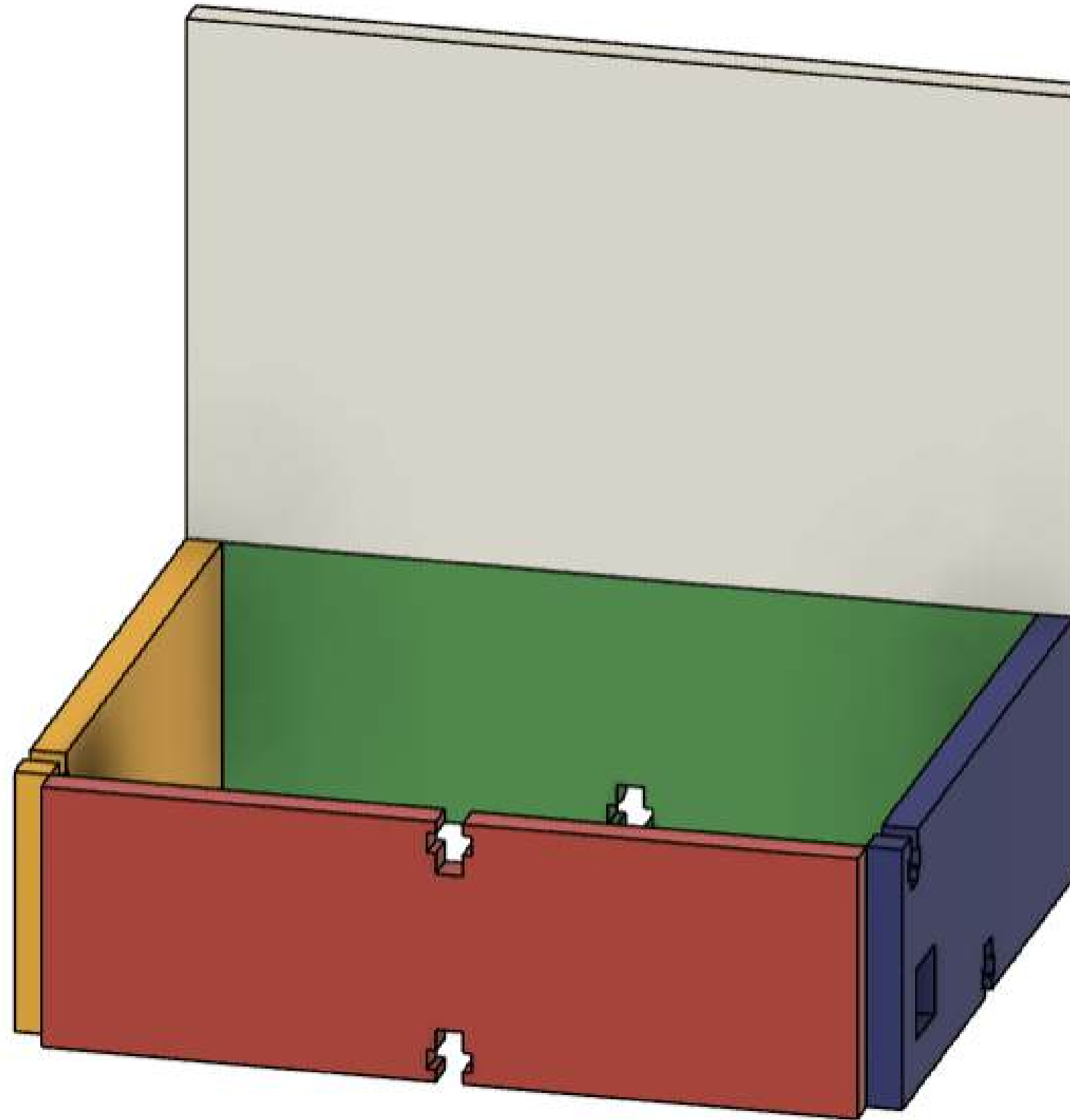
DESIGN BOX IN FUSION360: LEFT PANEL



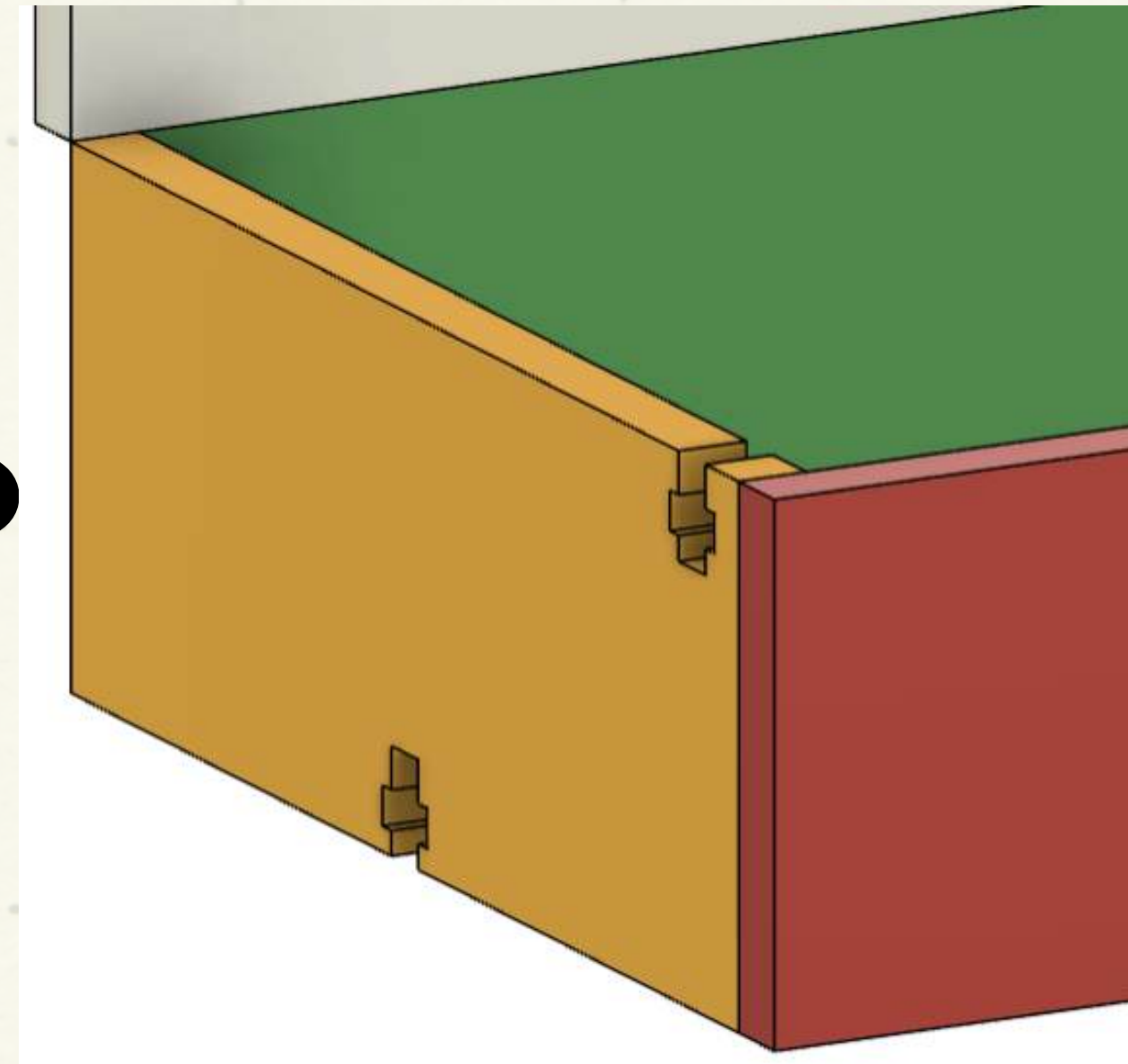
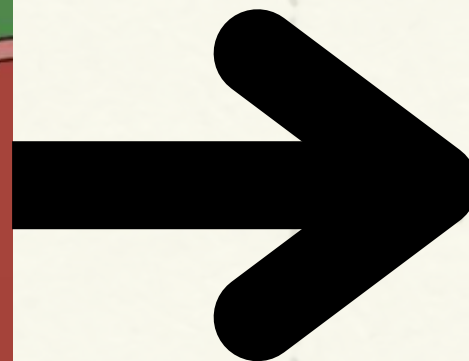
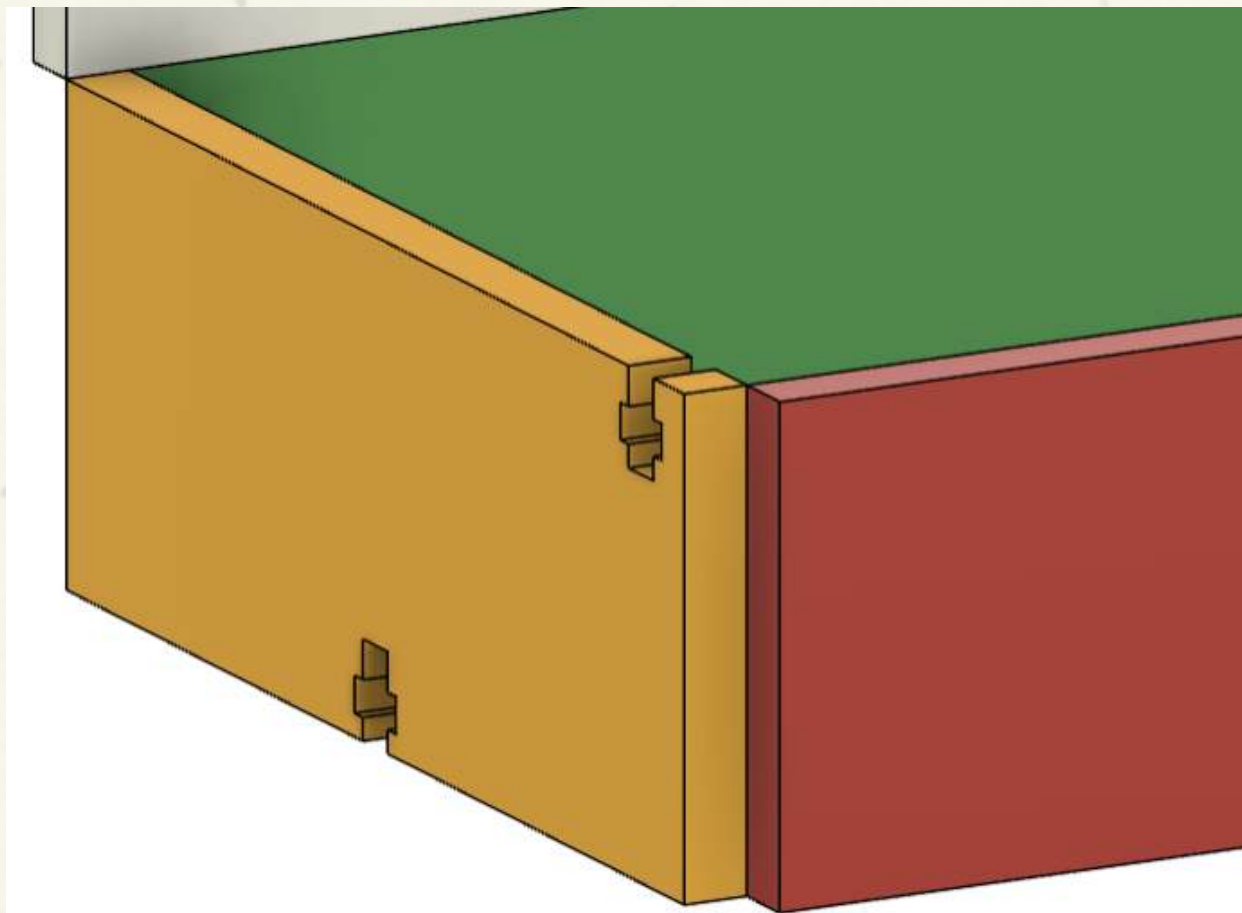
DESIGN BOX IN FUSION360: RIGHT PANEL



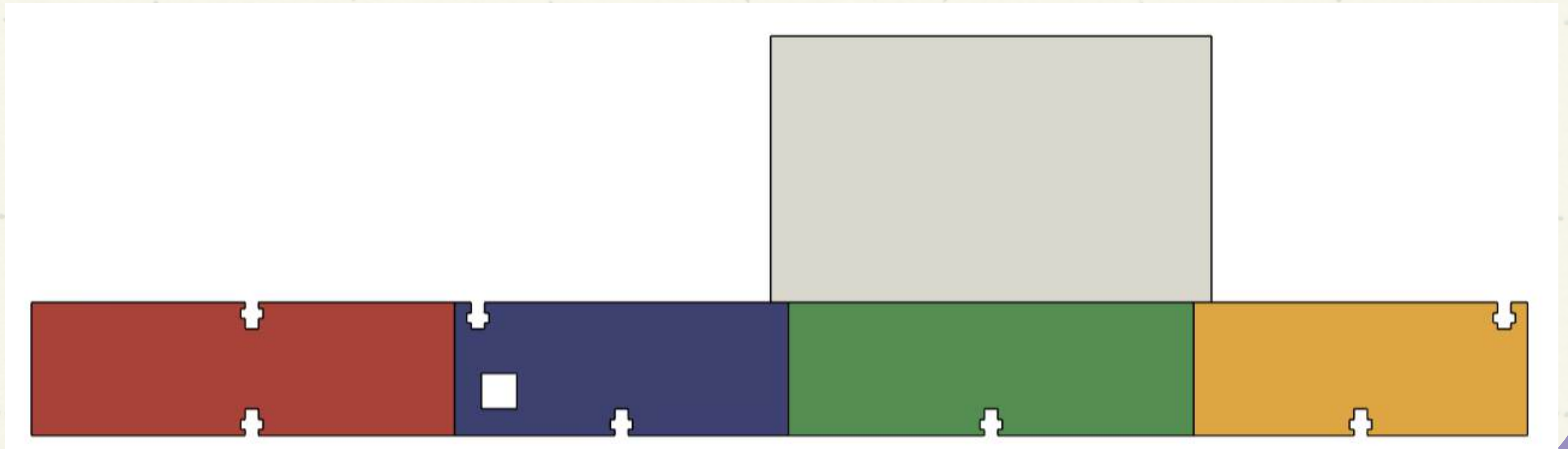
DESIGN BOX IN FUSION360: FULL BOX IN 3D



DESIGN BOX IN FUSION360: EXTRUDE RIGHT SIDE OF FRONT PANEL TO CLOSE THE BOX

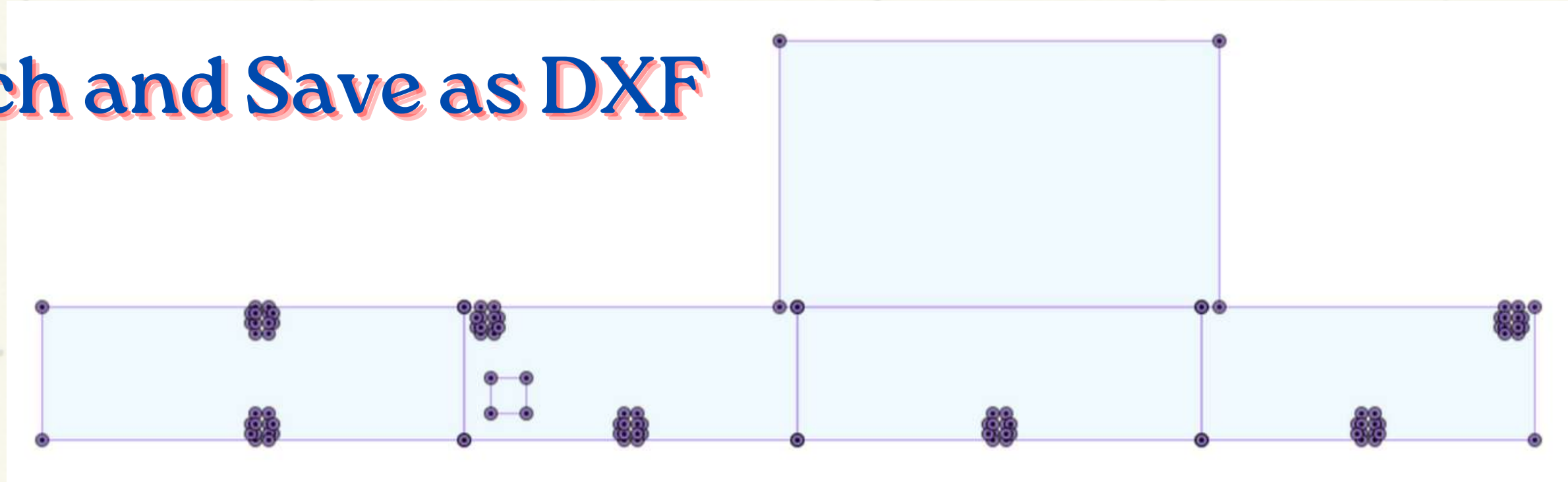


TO PUT THE BOX IN 2D, WE NEED TO 'UNWRAP' THE BOX. WE WILL DO THIS USING JOINT RELATIONSHIPS IN FUSION

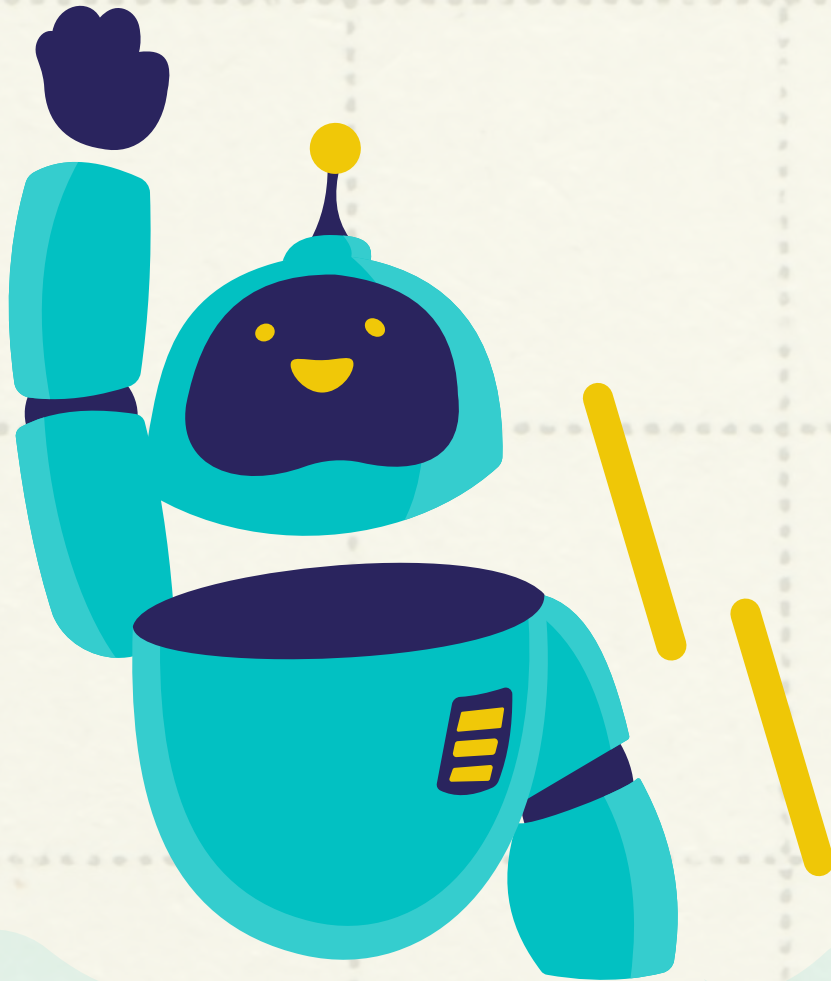


REMEMBER: DO NOT File->Export DXF

Right Click on the Sketch and Save as DXF



**Take it to Rhino to put things in the
Cut Layer and Score Layer**



Rhino Commands
Explode: Disconnect all line segments
SelDup: Select duplicate lines

LASER CUTTER



Check Settings to make sure the right material is selected and thickness is correct. Don't assume everyone is cutting cardboard!

Use material wisely. Try to minimize wasted material.

Stay by laser cutter to make sure there is no sustained fire.

Have a buddy!



4

Work on the Assignment:

Create a Box

Fusion Tutorials

3D Model 2 Objects

Cut Something on the Vinyl Cutter

